

Portfolio Audit & Fix Programme

Master brief + ready-to-paste stage prompts for the **portfolio-arena** site

Repository: ashish231309/portfolio-arena · branch arena/01a10c65-portfolio-arena

Base commit 85f12aa · deploy target Vercel · generated 2026-10-05

Status: **no changes applied yet** — the audit was read-only.

Part A is the context: project, findings, measured numbers, decisions, corrections and the image truth table. **Part B** holds one self-contained prompt per stage (S0-S10) — copy one block and paste it to start that stage.

Binding user corrections (these override any earlier suggestion): **C1** no third-party delivery note under the contact form · **C2** strip metadata from every file · **C3** present Full Stack Development as expertise, not learning.

Stage index:

S0 safety net · S1 images · S2 metadata · S3 404/deep links · S4 bug fixes · S5 accessibility · S6 performance · S7 discoverability · S8 positioning (Full Stack) · S9 hygiene/content/tests · S10 Tailwind 4

Contents

Part	Section	What it gives you
A	1-3 Project, ways of working, repository map	Identity, stack, routes, where every file lives
A	4-5 Audit method + findings register	Stable ids (A1-A4, T1-T35) referenced by the stage prompts
A	5.1 Change list in severity order	Issue -> fix -> files -> stage -> risk, four tiers
A	6 Measured values	Contrast table + verified replacement colours, asset inventory, WebP and PDF test results
A	7 USER CORRECTIONS C1-C3	Binding decisions that override earlier suggestions
A	8 IMAGE TRUTH TABLE	What each of the 16 screenshots really shows, new names, new alt text
A	9-11 Stage plan, open questions, maintenance	Ordering rules and what still needs your answer
B	Session opener	Paste this at the start of a brand-new chat
B	S0-S10 stage prompts	One copy-paste block per stage, each self-contained

PORTFOLIO MASTER BRIEF — Part A: Context & Decisions

What this file is. The complete, self-contained context for the `portfolio-arena` audit and fix programme.

If a chat session has no memory of earlier conversations, **paste this file (or attach it) plus the relevant stage prompt from** `PORTFOLIO-STAGE-PROMPTS.md`, and the work can continue exactly where it left off.

Companion file: `PORTFOLIO-STAGE-PROMPTS.md` (Part B) — one ready-to-paste prompt per stage.

Nothing in this programme has been applied yet. The repository was read-only throughout the audit.

Where these files live: `portfolio-arena/audit/` — untracked in git. Nothing was committed. To remove them completely: `rm -rf audit` from the repo root. Files stored outside the repository do not survive between sessions; files inside it do.

1. Project identity

Item	Value
Project	Personal portfolio of Ashish Kumar , B.Tech CSE student (Kanpur Institute of Technology, AKTU), graduating 2027
Positioning	Computer Science student building with Software, Web & Generative AI
Repo	<code>ashish231309/portfolio-arena</code> (GitHub), working copy at <code>/home/user/portfolio-arena</code>
Branch for all work	<code>arena/01a10c65-portfolio-arena</code> (session-bound — never switch, never push elsewhere)
Base commit at audit time	<code>85f12aa</code> ("Add files via upload"), 77 tracked files, working tree clean
Deploy target (decided)	Vercel
Art direction	"Signal & Ink" — editorial typography on warm ivory, deep-ink contrast fields, violet→cyan brand axis, coral/lime sparks. Full system in <code>DESIGN.md</code>
Licence	MIT

Stack (exact versions from `package-lock.json`)

- React **19.3.0** + Vite **8.3.0** (JavaScript/JSX — no TypeScript)
- Tailwind CSS **3.4.19** (custom token layer in `tailwind.config.js`)
- `motion` (Motion for React) **12.43.0** — springs, scroll-linked animation, route transitions
- `lenis` **1.3.26** — inertial smooth scrolling
- `react-router-dom` **7.18.4** · `lucide-react` **0.525.0**
- `@fontsource-variable/*` 5.x — self-hosted Space Grotesk (display), Manrope (body), JetBrains Mono (micro)
- Dev: `@vitejs/plugin-react` 5.2.0, `postcss`, `autoprefixer`, `playwright` 1.63.0 (unused for tests today)

Scripts

```
npm install / npm ci      # deps (NOTE: repo has no .gitignore until stage S0)
npm run dev               # vite --host 0.0.0.0 --port 5173
npm run build              # vite build -> dist/
npm run preview            # vite preview --host 0.0.0.0 --port 4173
```

`vite.config.js` already binds `0.0.0.0` with `allowedHosts: true` for both `server` and `preview`, so live previews work out of the box.

2. How work is done in this programme

1. **Batched delivery.** One stage per turn (S0...S10). Never mix two stages in one turn unless the user explicitly asks.
2. **Nothing without a command.** Analyse freely; change nothing until the user says go.
3. **Branch discipline.** All commits go to [arena/01a10c65-portfolio-arena](#). Pushing only to that branch. Pull requests are opened from it.
4. **Commit messages** are labelled with the stage id, e.g. `S1: fix swapped project screenshots, rename, re-point gallery + alt text`.
5. **Every stage ends with a live preview** (start the dev server with the process tools, bound to `0.0.0.0`) so the user can look at the result in their own browser.
6. **Report format** required at the end of every stage:
 - What changed (file-by-file, with counts)
 - What I verified, and how (build output, static render, measured numbers, DOM checks)
 - What I could **not** verify (and why)
 - Anything I need the user to decide
7. **Honesty in reporting.** If something is unverified, say so. Do not claim a visual check that was not performed.

Environment constraints (re-check at the start of any session)

- **Sandbox persistence:** the environment is rebuilt between sessions. Only files inside the git repository (`/home/user/portfolio-arena`) survive; `/tmp` and other paths outside the repo are wiped. Keep deliverables in `audit/` (untracked) and never rely on `/tmp` for anything that must last.
- **No browser for screenshots:** at audit time `playwright install chromium` failed, so screenshot-based visual QA was not possible. Verification was done instead via:
 - **production build** (`npx vite build`) — real bundle sizes and warnings;
 - **static render** of pages with `react-dom/server` bundled by esbuild — DOM structure, section/glow counts, heading order;
 - **measured contrast maths** (WCAG relative-luminance, including alpha compositing);
 - **binary inspection** of PDFs/images (metadata, dimensions);
 - `npm audit` (registry reachable at audit time).
- If a future environment does have a browser, prefer real screenshots — and if a change cannot be visually verified either way, say so explicitly and ask the user to look at the live preview.

3. Repository map

index.html	SEO/meta + JSON-LD Person block + #root
vite.config.js	server/preview host + allowedHosts
tailwind.config.js	colours, fonts, radii, shadows, keyframes <- design tokens
postcss.config.js	tailwindcss + autoprefixer
src/main.jsx	fonts + index.css + BrowserRouter -> App
src/App.jsx	Lenis setup, route transitions, shell (Intro/ScrollProgress/Cursor/Nav/Footer), noise layer
src/index.css	base layer, utilities (grid/dots/noise/underline/spotlight/mask), reduced-motion block
src/lib/site.jsx	SiteProvider (pointer MotionValues, tone, accent, cursor mode) + useSectionSignal
src/lib/motion.js	EASE, SPRING*, stagger/child variants, ACCENTS, accentText, accentBg
src/hooks/useMediaQuery.js	useMediaQuery, useFinePointer, useReducedMotionPref
src/hooks/usePageMeta.js	per-route document.title + meta description
src/components/	Section, SectionHead, Reveal (MaskLines/WordReveal/Fade/Stagger/Parallax/ScaleIn), Magnetic, Cursor, AmbientGlow, Intro, Marquee, Nav, Footer,
ScrollProgress, PageIntro	
src/sections/	Hero, About, Skills, ProjectsHome, Experience, Education, Certifications, Achievements, Contact <- each can render "bare" (without its own SectionHead)
src/pages/	Home, AboutPage, ProjectsPage, ProjectDetail, SimplePages (5 wrapper pages), NotFound
src/data/	profile, skills, projects, experience, education, certifications, achievements <- ALL content
public/	resume.pdf, favicon.svg, certificates/ (7 PDFs), projects/ (16 screenshots)
qa/shoot.mjs	hand-rolled Playwright screenshot script (not wired to npm; hardcodes /home/user/qa + localhost:5173)
DESIGN.md	design system, source of truth (v2.0)
README.md	overview, routes, where data/assets live, honesty policy
UPDATE_GUIDE.md	how to edit content safely + pre-publish checklist
audit/	THIS DOCUMENTATION SET (untracked; safe to delete)

Routes: / · /about · /projects · /projects/:slug (coding-ninjas, bmw) · /experience · /education · /certifications · /achievements · /contact · * (NotFound)

4. Audit method & raw evidence

Check	Command / method	Result	
Dependency advisories	<code>npm audit --package-lock-only</code>	5 × high: braces, chokidar, micromatch, fast-glob, tailwindcss	
Same, shipped tree only	<code>npm audit --omit=dev</code>	0 vulnerabilities (18 prod deps)	
Secrets in code	grep for keys/tokens/passwords	none — only the public Gmail in the FormSubmit endpoint	
XSS sinks	grep <code>dangerouslySetInnerHTML</code> , <code>eval</code> , <code>innerHTML</code> , <code>document.write</code>	none	
External links	grep <code>target="_blank"</code>	all carry <code>rel="noreferrer"</code>	
Storage/tracking	grep <code>localStorage</code> , <code>cookie</code>	only one <code>sessionStorage</code> flag for the intro overlay; no analytics, no cookies	
Image metadata	EXIF/GPS/XMP extraction over all 16 files	clean (no GPS, no device, no editor traces)	

Check	Command / method	Result	
PDF metadata	Stream/Info/XMP extraction over all 7 PDFs	resume: Author "Un-named", Word 2010, created 14 Sep 2026 (text also contains phone +91 85459 34143); YSF: Author "Richa Garg" (third party) + Canva; others: tool names only (Qt 5.5.1, PDFsharp 6.1.1, plpdf 2.8.0, Adobe Illustrator 28.5)	
Production build	<code>npm vite build</code> in a throwaway copy	JS 519.81 kB (gzip 160.26) single file + Vite ">500 kB" warning; CSS 50.39 kB (gzip 14.44); 14 woff2 = 214 kB; dist = 12 MB	
DOM structure	static render of 7 routes	2 <code><Section></code> s × 2 ambient glows on 5 sub-pages; heading order h1→h3 (no h2) on 5 routes	
SSR safety	render <code>Hero</code> outside a browser	throws <code>ReferenceError: window is not defined</code>	
Contrast	WCAG luminance maths incl. alpha compositing	5 AA failures (see §6)	
Lenis behaviour	read <code>node_modules/lenis/dist/lenis.mjs</code>	<code>onNativeScroll</code> only re-syncs when <code>isScrolling === false</code> \	'native' → <code>native scrollTo</code> during a smooth scroll is ignored

5. Findings register (IDs are stable — stage prompts reference them)

Security / privacy

ID	Finding
A1	5 high dev-only advisories (Tailwind 3 tooling chain). Shipped tree clean. Only fix = Tailwind 4 (breaking) → stage S10
A2	Contact form: recipient Gmail is in the public bundle (https://formsubmit.co/ajax/ashish1492a@gmail.com); no request timeout → button can hang on "Sending..."
A3	PDF metadata (see §7). User decision C2: strip metadata from EVERY file → stage S2
A4	No error boundary: any runtime error or failed JS chunk = blank white page → stage S0

Bugs (visitor-visible)

ID	Finding
T14	Navbar can turn invisible. <code>useSectionSignal</code> sets tone once when a section enters the middle band and never restores. Sub-pages nest a <code><Section></code> inside another <code><Section></code> (2 sections, 2 glows) → on <code>/contact</code> and <code>/certifications</code> a sticky dark tone over an ivory background = unreadable nav links. Proven by static render (<code>/contact sections=2 glows=2, /certifications sections=2 glows=2, /about glows=2</code>) → stage S4
T15	Two scroll engines: route change uses native <code>window.scrollTo</code> , footer uses native smooth scroll, while Lenis owns scrolling → jumps/rubber-banding when navigating during a scroll → stage S4
T16	Mobile menu only sets <code>body { overflow:hidden }</code> ; Lenis keeps scrolling behind it; focus can Tab out of the overlay → stage S4
T17	Cursor promises clicks that don't exist: <code>data-cursor="cert"/"project"</code> on non-clickable cards and the project mockup → stage S4
T18	Hero orbit tilt reads <code>window.innerHeight/innerWidth</code> during render; freezes on resize and breaks SSR/prerender → stage S4
T19	Contact form: no timeout (see A2), error copy says "email me directly below" while the address is in the left column → stage S4. Constraint C1: do NOT add a third-party delivery note

ID	Finding
T10	Featured-project GitHub button renders <code>href={null}</code> if the featured project has no repo → stage S6

Accessibility (measured — see §6 for exact numbers)

ID	Finding
T25	C1 focus ring cyan on ivory = 1.61:1 (invisible on light sections); C2 <code>muted</code> on ivory/tint = 4.45/4.22; C3 <code>text-cobalt</code> small labels on ivory = 3.29; C4 ivory on indigo CTA = 4.35; C5 form input borders <code>ivory/20</code> = 1.81 + placeholder <code>fog/50</code> = 2.67; C6 teal chip 4.07; C7 <code>muted/80</code> micro notes 3.10 → stage S5
T26	Heading order skips h2 on <code>/experience</code> , <code>/education</code> , <code>/certifications</code> , <code>/achievements</code> , <code>/contact</code> → stage S5
T5	Marquee first row is not <code>aria-hidden</code> → screen readers read it twice → stage S5
T7	<code>/achievements</code> page header uses coral while its section is lime (breaks the one-accent rule) → stage S5
T12	Unguarded lookups: <code>TILE[item.id]</code> / <code>CHIP[item.id]</code> (achievements), <code>accentText[...]</code> — a new data entry can break the mosaic → stage S5
T3	No <code>MotionConfig</code> ; reveals/parallax/tilt ignore <code>prefers-reduced-motion</code> despite <code>DESIGN.md §8</code> → stage S0

Images (verified by opening all 16 files)

ID	Finding
T24	Both project folders are crossed. Everything in <code>public/projects/bmw/</code> is really Coding Ninjas ; everything in <code>public/projects/coding-ninjas/</code> is really BMW . Alt text describes the intended project, so the site shows BMW screenshots captioned "Coding Ninjas recreation". Two images (<code>bmw-08</code> , <code>bmw-09</code>) are attached nowhere. Names are opaque (<code>cn-01</code> , <code>bmw-04</code>). Full mapping + new names + new alt text in §8 → stage S1

Performance / SEO / hygiene

ID	Finding
T31	519.81 kB single JS bundle (160.26 kB gzip); all 10 routes downloaded before first paint → stage S6
T32	No WebP/ <code>srcset</code> ; <code>dist</code> 12 MB; largest objects: YSF cert 4.43 MB, <code>cn-04</code> 1.10 MB, <code>cn-01</code> 743 kB, <code>resume</code> 647 kB, Oracle cert 524 kB; 14 font files shipped (latin subsets ≈87 kB) → stage S6
T30	Gallery images lack <code>width/height</code> → layout shift; the ~2.2:1 screenshots sit in an <code>aspect-[16/10]</code> box with <code>object-cover object-top</code> → side cropping → stage S6
T27	No <code>og:image</code> /twitter image/canonical (blank share preview); Home calls <code>usePageMeta(null, ...)</code> which overwrites the rich static title/description from <code>index.html</code> → stage S7
T28	No <code>robots.txt</code> , no <code>sitemap.xml</code> → stage S7
T29	No web manifest / Apple touch icon → stage S7
T20	Gmail address in the public bundle (same root cause as A2) → stage S7 (bundle side) with S4 handling the form behaviour
T21	No host-level 404: no <code>404.html</code> , and the React <code>NotFound</code> only covers in-app routes → stage S3
T22	No SPA rewrite for Vercel → <code>yoursite.com/projects/bmw</code> on a direct load/refresh 404s → stage S3
T1	No .gitignore — <code>node_modules/</code> , <code>dist/</code> , <code>qa/*.png</code> are all unignored → stage S0
T2	No error boundary (same as A4) → stage S0
T4	Dead imports: <code>useSite</code> (Section.jsx), <code>Terminal/Stagger/StaggerItem</code> (About.jsx), <code>Stagger/StaggerItem</code> (Education.jsx), <code>EASE</code> (Hero.jsx), <code>ExternalLink/MaskLines</code> (ProjectsHome.jsx), <code>Parallax</code> (ProjectDetail.jsx); unused <code>labelledBy</code> prop on <code>Section</code> → stage S0
T11	Hardcoded copy that goes stale: "Two recreations" (ProjectsPage), "Project 02" (ProjectsHome), <code>FRAME_LABELS</code> (7 fixed captions) → stages S1/S9

ID	Finding
T13	DESIGN.md states wrong numbers: muted contrast "5.0:1" (really 4.45), indigo "4.6:1" (really 4.35); claims Lenis pauses when the tab is hidden (not implemented) → stage S9
T6	Badminton achievement has no year while every other entry does → stage S9 (needs the real year from the user)
T8	Cursor rail label SWIPE → vs DESIGN.md §7 SCROLL → stage S9
T23	qa/shoot.mjs not wired to npm, hardcodes <code>/home/user/qa/</code> + <code>localhost:5173</code> , and its overflow check is defeated by <code>body { overflow-x: clip }</code> → stage S9
T33	No ESLint/Prettier, no lint script → stage S9
T34	No automated tests; no check that a project shows its own screenshots → stage S9
T35	Tailwind 3 → 4 (breaking; the only way to clear A1) → stage S10
T9	CV headline says "Full Stack Development" while the site never claims it → resolved by C3 in favour of expertise → stage S8
F-corrections	Two earlier mis-statements, retracted: (a) "duplicate form IDs" was a false alarm (<code>id="cf-name"</code> appears once as a prop and once as the real input attribute → one DOM id); (b) the tone bug bites only on /contact and /certifications (the other three sub-pages nest matching colours; they still render a double glow)

5.1 Change list in severity order (issue → fix → files → stage → risk)

Tier 1 — tiny, near-zero risk

ID	Issue	Fix	Files	Stage	Risk
T1	No .gitignore	Standard ignore file (node_modules, dist, qa screenshots, .env*)	.gitignore (new)	S0	none
T2	Runtime error = blank page	ErrorBoundary + branded fallback, Nav/Footer stay alive	components/ErrorBoundary.jsx (new), App.jsx	S0	none
T3	Reduced-motion ignored	<code><MotionConfig reducedMotion="user"></code> wrapper	App.jsx	S0	low
T4	8 dead imports + unused prop	Delete them	Section, About, Education, Hero, ProjectsHome, ProjectDetail	S0	none
T5	Marquee read twice by AT	Hide both visual rows, expose text once	Marquee.jsx	S5	none
T6	Badminton has no year	Add the year (user supplies it)	data/achievements.js	S9	none
T7	/achievements accent mismatch	Align page-intro accent with the section	pages/SimplePages.jsx	S5	low
T8	Cursor label vs design doc	Make code and doc agree	Cursor.jsx or DESIGN.md	S9	none
T10	<code>href={null}</code> GitHub button	Guard / hide	ProjectsHome.jsx	S6	none
T11	"Two recreations", "Project 02" hardcoded	Derive from data	ProjectsPage, ProjectsHome	S9	none
T12	Unguarded <code>TILE[id]/accent</code> lookups	Safe fallbacks	Achievements, Skills, Certifications, SectionHead	S5	none
T13	DESIGN.md wrong numbers + Lenis claim	Correct doc / implement the pause	DESIGN.md	S9	none

Tier 2 — small, contained logic

ID	Issue	Fix	Files	Stage	Risk
T14	Navbar turns invisible; double section + glow on 5 sub-pages	intro slot on Section, tone restore in useSectionSignal , single section per page	lib/site.jsx , Section.jsx , 6 pages, touched sections	S4	medium
T15	Two scroll engines	One Lenis instance via context; route reset + footer through it; resize() after route	lib/scroll.jsx (new), App, Footer	S4	medium
T16	Mobile menu doesn't lock scroll / focus escapes	lenis stop/start, dialog semantics, focus trap, inert background, focus restore	Nav.jsx	S4	low
T17	Cursor promises non-existent clicks	Real links on cards/mockup, or downgrade cursor state	ProjectsHome, Certifications, Experience	S4	low
T18	Hero freezes on resize, breaks SSR	useViewport hook instead of render-time window reads	hooks/useViewport.js (new), Hero.jsx	S4	low
T19	Form: no timeout + wrong error copy	AbortController 10 s; fix copy (no third-party note — C1)	Contact.jsx	S4	low
T20	Gmail in public bundle	Env-driven alias endpoint; verify by grepping the build	Contact.jsx , .env.example	S7	low
T21	No host 404	public/404.html , no-JS, branded, design-consistent	public/404.html (new)	S3	none
T22	Deep links 404 on Vercel	vercel.json SPA rewrite	vercel.json (new)	S3	low
T23	QA script unwired + broken overflow check	npm script, repo output path, measure body.scrollWidth , add BMW page	qa/shoot.mjs , package.json	S9	none

Tier 3 — medium (images, accessibility, discoverability, performance)

ID	Issue	Fix	Files	Stage	Risk
T24	All 16 screenshots in the wrong folder, 2 orphans, opaque names	Move + rename per §8.2, new alt text §8.3, data-driven captions, attach orphans	16 images, data/projects.js , ProjectsHome.jsx , README, UPDATE_GUIDE	S1	low
T25	5 measured AA contrast failures	Apply §6.2 values verbatim	tailwind config, index.css, Hero, Nav, Contact, Experience, Skills, Achievements	S5	medium
T26	Heading order skips h2 on 5 routes	Render section heading or demote entry titles	SectionHead.jsx , sections, SimplePages	S5	low
T27	No social preview; Home clobbers meta	og:image + twitter + canonical; keep static meta on /	index.html , public/og-cover.png , Home/usePageMeta	S7	low
T28	No robots/sitemap	Add both with the 9 routes	public/robots.txt , public/sitemap.xml	S7	none
T29	No manifest / apple icon	Manifest + 180×180 icon + links	index.html , public/site.webmanifest	S7	none
T30	No image dimensions; wide shots cropped	width/height + optional aspect retune (ask)	data/projects.js , ProjectDetail, ProjectsHome	S6	low
T31	519.81 kB single JS bundle	Route-level lazy + Suspense, build config	App.jsx , vite.config.js	S6	medium
T32	No WebP/srcset; 12 MB deploy; 14 font files	Lossless WebP for the 7 PNG-backed files only, srcset, latin-only fonts, PDF image question	images, main.jsx , index.html	S6	medium

ID	Issue	Fix	Files	Stage	Risk
T33	No linter/formatter	ESLint (react-hooks) + Prettier + scripts	config files, package.json	S9	low
T34	No tests	Playwright smoke tests incl. "each project shows its own screenshots"	tests/smoke.spec.js	S9	low
C2/A3	Metadata on 7 PDFs (2 with personal data) + whole sweep	Strip from every PDF/image/SVG + check:meta verifier	all of public/ , qa/check-metadata.*	S2	low
C3/T9	Full Stack not claimed; CV contradicts the site	Expertise skill group from existing technologies + doc alignment (no invented stack)	data/skills.js , data/profile.js , About/Skills, README, UPDATE_GUIDE	S8	low

Tier 4 — large / breaking

ID	Issue	Fix	Files	Stage	Risk
T35	5 high dev-only advisories (Tailwind 3 chain)	Tailwind 3 → 4, config layer only, isolated, last	index.css , tailwind.config.js , postcss.config.js , package.json	S10	high

6. Measured reference values

6.1 Contrast (WCAG AA: 4.5:1 normal text, 3:1 large text / UI)

Pair	Ratio	Verdict
ink on ivory	16.03	pass
ivory on deep / ink / deeper	15.51 / 16.03 / 16.84	pass
fog on deep / deeper / ink	6.84 / 7.42 / 7.07	pass
muted on ivory	4.45	fail
muted on tint #EFEAF9	4.22	fail
muted on paper	4.86	pass
indigo on ivory	4.35	fail
cobalt #3B82F6 on ivory	3.29	fail
cobalt on deep	4.71	pass
cyan #27D3F2 on ivory (focus ring)	1.61	fail
cyan on deep / ink	9.65 / 9.97	pass
coral on ivory	2.48	fail (icons only)
teal #0e7d90 on cyan/10-over-ivory	4.07	fail
muted/80 on ivory	3.10	fail
form border ivory/20 on #111A2C	1.81	fail
placeholder fog/50 on #111A2C	2.67	fail
ivory on indigo CTA	4.35	fail (borderline)

6.2 Verified replacement values (computed, use these — do not re-guess)

Purpose	Change	New ratio
Secondary text token	muted #667085 → #5B6474	5.34 ivory · 5.06 tint · 5.82 paper

Purpose	Change	New ratio
Small accent text on light	add cobalt-ink #1D4ED8 (keep bright cobalt on dark)	5.99 on ivory
Primary button background	#6C5CE7 → #5A4BD4 (already the hover colour)	ivory on it = 5.54
Focus ring, light sections	cyan → ink #10162B (cyan stays on dark zones)	16.03
Form input border	ivory/20 → ivory/45	4.10
Form placeholder	fog/50 → fog/80	4.88
Achievements teal chip	#0e7d90 → #0b6b7c	5.20
Micro mono notes	muted/80 → full muted	5.34

6.3 Asset inventory

Coding Ninjas screenshots (currently in `public/projects/bmw/`, all `.jpeg`) `bmw-01` 1259×716 · `bmw-02` 1264×711 · `bmw-03` 1266×714 · `bmw-04` 1260×712 · `bmw-05` 1264×717 · `bmw-06` 1279×713 · `bmw-07` 1265×711 · `bmw-08` 1277×707 · `bmw-09` 1279×711 — 108/94/175/111/85/100/103/93/87 kB (956 kB total)

BMW screenshots (currently in `public/projects/coding-ninjas/`, all `.png`) `cn-01` 1349×613 (743 kB) · `cn-02` 1351×613 (471 kB) · `cn-03` 1351×613 (71 kB) · `cn-04` 1351×620 (1105 kB) · `cn-05` 1352×620 (50 kB) · `cn-06` 1349×610 (125 kB) · `cn-07` 1348×607 (443 kB) — 3008 kB total

Other assets: `resume.pdf` 647 kB · 7 certificate PDFs (YSF 4.43 MB, Oracle 524 kB, Gemini Educator 451 kB, Walmart 279 kB, Digital Productivity 216 kB, Deloitte 120 kB, Gemini Student 57 kB) · `favicon.svg` 471 B

6.4 WebP conversion, pre-tested (`libwebp lossless=True, quality=100, method=5`)

Group	Files	Today	Lossless WebP	Change
BMW project (the <code>.png</code> files)	7	3008 kB	1545 kB	−49% (pixel-identical)
Coding Ninjas project (the <code>.jpeg</code> files)	9	956 kB	1868 kB	+95% — lossless is counter-productive

Lossy q85 for reference: total drops to 818 kB (−79%), but that is **lossy** and needs the user's explicit approval. Decision recorded: **lossless WebP only for the PNG-backed 7**; for the 9 JPEG-backed files the default is **leave as JPEG** (optional lossy WebP q88–92 later, only on request).

6.5 PDF metadata removal — tested on copies

File	Removed	Integrity proof	Size
<code>resume.pdf</code>	Info dict + XMP (Author "Un-named", Word 2010, dates)	1 page → 1 page, content-stream sha256 identical	647 → 644 kB
<code>ysf-internship-certificate.pdf</code>	Info + XMP (Author "Richa Garg")	1 page → 1 page, content-stream sha256 identical	4433 → 4434 kB

Method: `pikepdf — open_metadata().clear()`, delete `docinfo`, re-save with `compress_streams=True`. Nothing visible changes. The YSF file stays 4.4 MB because its bulk is embedded imagery, not metadata (shrinking it needs image re-compression inside the PDF — stage S6, with a legibility comparison first).

7. USER CORRECTIONS — binding, they override anything written elsewhere

These three items are the user's explicit decisions. Where this document or a stage prompt contradicts them, **the correction wins**.

C1 — Do NOT add a FormSubmit disclosure line

Earlier proposal A2/T19 suggested adding "Messages are delivered via FormSubmit (third party)" under the contact form. **The user does not want that line. Do not add it.** Still allowed in the same area: the 10-second request timeout, and fixing the error copy that says "email me directly below" (the address sits in the left column).

C2 — Remove metadata from EVERY file in the portfolio

Not just the two PDFs found in the audit. **All 7 PDFs, all 16 screenshots, and `favicon.svg`** must end up metadata-free, and this must be verified file-by-file afterwards. Where a file had no metadata, record that fact ("verified clean") rather than skipping it. Method per type:

- **PDF** → `pikepdf` (as §6.5)
- **PNG** → lossless re-save (drop ancillary chunks: `tEXt/iTXt/zTXt/tIME/eXIf`; pixels byte-identical)
- **JPEG** → re-save with `quality='keep'`, `subsampling='keep'` so the image data is **not** recompressed, only metadata markers dropped (APPn/COM)
- **SVG** → strip XML comments/metadata nodes; keep the drawing untouched
- **WebP** (where produced later) → save without EXIF/XMP chunks

Deliverable of that stage: a verifier script (e.g. `qa/check-metadata.mjs` or a Python equivalent) that prints a per-file verdict and **fails** if any metadata is found. This becomes part of `npm run check:meta`.

C3 — Present Full Stack Development as EXPERTISE, not as learning

The site currently never claims full-stack; `README.md`'s honesty policy even says learning topics are "shown as exploring, never as production expertise", while the CV headline lists "Full Stack Development". The user wants the **site to claim Full Stack Development as an expertise area**. Implementation:

- Add a **Full Stack Development** skill group in `src/data/skills.js` built only from technologies already present in the data (frontend: HTML/CSS/JavaScript/React/Tailwind/Vite; server-side & data: Python/Java/SQL/DBMS). Do **not** invent frameworks, years of experience or metrics the user has not supplied.
- Renumber the skill group indexes cleanly (they are displayed as `01...07`).
- Reflect it in the About copy and keep the CV aligned (the CV already says full-stack, so **no CV edit is needed** — this also resolves T9 in favour of expertise).
- Update the honesty-policy wording in `README.md` + `UPDATE_GUIDE.md` so the docs no longer contradict the site. **Keep** the third-party protections untouched: "Website Recreation" labels, "Virtual Job Simulation — not employment", and "no affiliation/endorsement implied" must stay exactly as they are — those are about other companies, not about self-assessment.
- The `currentlyExploring` chips for infrastructure topics (Linux, Docker, hosting, APIs) stay as they are **unless** the user later says otherwise.
- Open detail the user must answer: whether a specific backend framework should be named (Node/Express/Django/Flask) — do not guess.

8. IMAGE TRUTH TABLE — verified by opening all 16 files

Root cause: the two screenshot batches were uploaded into each other's folder, and `src/data/projects.js` alt text was written for the intended project. Result: mismatched captions, two orphaned images, opaque file names.

8.1 What each file actually shows

Current path	Actually shows	Belongs to
<code>bmw/bmw-01.jpeg</code>	CN hero — "Give your career an unfair AI advantage" + AI-tool marquee	Coding Ninjas
<code>bmw/bmw-02.jpeg</code>	CN — "Let's find the right course for you" finder + placement-stats row	Coding Ninjas

Current path	Actually shows	Belongs to
bmw/bmw-03.jpeg	CN — "Pick the track that matches where you are" course rails + filter chips	Coding Ninjas
bmw/bmw-04.jpeg	CN — "AI infused curriculum curated by experts" + student quote card	Coding Ninjas
bmw/bmw-05.jpeg	CN — testimonial carousel, reviewer tabs (URL localhost:5173/#/reviews)	Coding Ninjas
bmw/bmw-06.jpeg	CN — "What the industry is saying" press cards	Coding Ninjas
bmw/bmw-07.jpeg	CN — 10X Club community section + mentor rail	Coding Ninjas
bmw/bmw-08.jpeg	CN — "1,50,000+ alumni" ratings band + NSDC accreditation strip	Coding Ninjas — orphaned
bmw/bmw-09.jpeg	CN — full footer (incl. "Non-commercial learning project — React + Tailwind CSS recreation")	Coding Ninjas — orphaned
coding-ninjas/cn-01.png	BMW — "THE 5 LONG WHEELBASE" hero + "Discover now"	BMW
coding-ninjas/cn-02.png	BMW — dealer locator map page	BMW
coding-ninjas/cn-03.png	BMW — search page + quick-links/legal footer	BMW
coding-ninjas/cn-04.png	BMW — diagonal-split All Models showcase (1 Series, 5 Series, X1, i7)	BMW
coding-ninjas/cn-05.png	BMW — "BOOK A TEST DRIVE" request form	BMW
coding-ninjas/cn-06.png	BMW — all-models listing with price-range/instalment filters	BMW
coding-ninjas/cn-07.png	BMW — "BMW Special Offers" panel (EMI figure + Know More)	BMW

8.2 New file names (pattern <project-slug>-NN-<content>.<ext>)

→ [public/projects/coding-ninjas/](#) (kept as .jpeg)

New name	From	Gallery order
coding-ninjas-01-hero.jpeg	bmw/bmw-01.jpeg	1
coding-ninjas-02-course-finder.jpeg	bmw/bmw-02.jpeg	2
coding-ninjas-03-course-rails.jpeg	bmw/bmw-03.jpeg	3
coding-ninjas-04-ai-curriculum.jpeg	bmw/bmw-04.jpeg	4
coding-ninjas-05-10x-club.jpeg	bmw/bmw-07.jpeg	5
coding-ninjas-06-testimonials.jpeg	bmw/bmw-05.jpeg	6
coding-ninjas-07-alumni-ratings.jpeg	bmw/bmw-08.jpeg	7 — newly attached
coding-ninjas-08-in-the-news.jpeg	bmw/bmw-06.jpeg	8
coding-ninjas-09-footer.jpeg	bmw/bmw-09.jpeg	9 — newly attached

→ [public/projects/bmw/](#) (converted to .webp in stage S6; .png until then)

New name	From	Gallery order
bmw-01-hero-5-series.png	coding-ninjas/cn-01.png	1
bmw-02-all-models.png	coding-ninjas/cn-04.png	2
bmw-03-dealer-locator.png	coding-ninjas/cn-02.png	3
bmw-04-model-listing.png	coding-ninjas/cn-06.png	4

New name	From	Gallery order
bmw-05-special-offers.png	coding-ninjas/cn-07.png	5
bmw-06-test-drive-form.png	coding-ninjas/cn-05.png	6
bmw-07-search-footer.png	coding-ninjas/cn-03.png	7

8.3 New alt text (describes what the reader actually sees)

Coding Ninjas

1. Coding Ninjas recreation – hero with "Give your career an unfair AI advantage" headline and AI-tool marquee
2. Coding Ninjas recreation – "Let's find the right course for you" form panel with placement stats row below
3. Coding Ninjas recreation – "Pick the track that matches where you are" course rail with category filter chips
4. Coding Ninjas recreation – "AI infused curriculum curated by experts" feature list with student quote card
5. Coding Ninjas recreation – 10X Club community section with weekly mentor rail
6. Coding Ninjas recreation – student testimonial carousel with reviewer tabs
7. Coding Ninjas recreation – 1,50,000+ alumni ratings band and NSDC accreditation strip
8. Coding Ninjas recreation – "What the industry is saying" press coverage cards
9. Coding Ninjas recreation – multi-column footer with offerings, products, community and payment row

BMW

1. BMW recreation – 5 Series Long Wheelbase hero with "Discover now" call to action
2. BMW recreation – diagonal-split All Models showcase (1 Series, 5 Series, X1, i7)
3. BMW recreation – dealer locator map page with nearby-location search fields
4. BMW recreation – all-models listing with price-range and instalment sliders
5. BMW recreation – special offers panel with monthly EMI figure and finance links
6. BMW recreation – book a test drive request form with model and city selectors
7. BMW recreation – site search page with quick links and legal footer

8.4 Consequences to implement alongside the swap

- `src/data/projects.js`: rewrite both `gallery` arrays (paths + alt text), add a `label` per gallery item for the sticky-story captions, update `scrollFrames` for the 9-image CN gallery.
- `src/sections/ProjectsHome.jsx`: replace hardcoded `FRAME_LABELS` with `project.gallery.map(g => g.label)`.
- `README.md` asset section and `UPDATE_GUIDE.md` step 2: document the naming pattern + "one gallery array per project, in page order".
- `qa/shoot.mjs`: add the BMW detail page so this class of mix-up is caught (stage S9/T34).
- Aspect note: CN screenshots are ~2.2:1 but the sticky story box is `aspect-[16/10]` with `object-cover object-top` → sides get cropped. Optional retune (e.g. `aspect-[2/1]`) — ask the user.
- Visible-copyright note: some screenshots contain third-party contact details/footers. They are already labelled recreations; an optional extra line under galleries ("Unofficial study project — no affiliation with the brand shown") is proposed but **needs the user's OK** (it is not covered by C1, which is only about the form).

9. Stage plan (see Part B for the paste-ready prompts)

Stage	Name	Items	Notes
S0	Safety net	T1, T2, T3, T4	.gitignore must exist before any <code>npm install</code> . Almost zero visual risk
S1	Images truth pass	T24	Pure moves/renames + data edits. Biggest visible correctness win
S2	Metadata sweep (all files)	C2 → A3/T36 extended	7 PDFs + 16 images + favicon.svg, with a verifier script
S3	404 + deep links + failure surfaces	T21, T22 (+T2 404.html)	Vercel: vercel.json rewrite + public/404.html ; unify host 404 / React 404 / error fallback
S4	Bug fixes	T14, T15, T16, T17, T18, T19 (+A2 timeout)	The real broken behaviour. C1 applies (no delivery note)
S5	Accessibility	T25, T26, T5, T7, T12	Use §6.2 values verbatim
S6	Performance	T31, T32, T30, T10	Includes lossless WebP for the 7 PNG-backed files only
S7	Discoverability & privacy	T27, T28, T29, T20	Social card, robots/sitemap, manifest, remove the address from the bundle
S8	Positioning	C3	Full Stack Development as expertise + doc honesty-policy alignment
S9	Hygiene, content & tests	T6, T8, T11, T13, T23, T33, T34	Needs the badminton year from the user
S10	Tailwind 4 (isolated, last)	T35	Only after the rest is committed to main ; clears A1

Ordering rules: S0 first (it creates [.gitignore](#) and the error boundary that make everything else safer). S1 before S2 (names settle before metadata is stripped). S6 after S1/S2 (conversion depends on final names). S10 last and alone. Any stage may be skipped.

10. Open questions for the user

1. **Badminton year** (T6) — needed before S9; not invented.
2. **Backend stack for the Full Stack expertise claim** (C3) — may a specific framework be named (Node/Express, Django, Flask) or should the claim stay at the technologies already listed?
3. **JPEG-backed screenshots** — leave as JPEG (default), or lossy WebP q88-92 for ~60% smaller files?
4. **Sticky-story aspect ratio** for the wide CN screenshots — retune or leave?
5. **Extra "no affiliation" line under galleries** — yes/no (separate from C1).
6. **The user's own fix ideas** — send them before the stage they touch so they are built in rather than reworked.

11. Document maintenance

If a stage changes a decision recorded here (e.g. the user picks lossy WebP, or a finding turns out to be wrong), **update this file in the same turn**. Part B prompts reference this document by item id (T-codes, S-stages, C-corrections) so the two files must stay consistent.

Persistence rule: keep these files in [portfolio-arena/audit/](#). Only files inside the git repository survive a session rebuild; anything in `/tmp` or other locations is discarded.

End of Part A.

PORTFOLIO STAGE PROMPTS — Part B: ready to paste

How to use this file. Each stage below is a self-contained prompt inside a copy-paste block. Copy **one block** and paste it into the chat. That block alone carries enough context for the work to run correctly even with no memory of earlier conversations.

For extra safety, paste (or attach) **Part A — PORTFOLIO-MASTER-BRIEF.md** at the start of a session; every prompt refers to its item ids (T-codes, S-stages, C-corrections).

Stages are independent and may be run in any order, except: **S0 first** and **S10 last**. One stage per turn.

Stage index: S0 safety net · S1 images · S2 metadata · S3 404/deep links · S4 bug fixes · S5 accessibility · S6 performance · S7 discoverability · S8 positioning (Full Stack) · S9 hygiene/content/tests · S10 Tailwind 4

Files in this set: [audit/PORTFOLIO-MASTER-BRIEF.md](#) (Part A) · [audit/PORTFOLIO-STAGE-PROMPTS.md](#) (this file) · [audit/PORTFOLIO-AUDIT-AND-FIX-PLAN.pdf](#) (both, printable).

SESSION OPENER (optional — paste before the stage prompt when starting fresh)

You are continuing an audit-and-fix programme on my portfolio project. Attached/pasted is PORTFOLIO-MASTER-BRIEF.md (Part A) which contains the full context, all findings, decisions and constraints. Read it first. Then I will paste one stage prompt (from PORTFOLIO-STAGE-PROMPTS.md, Part B) and you will execute only that stage.

Ground rules for every stage:

- Work only in /home/user/portfolio-arena, branch arena/01a10c65-portfolio-arena. Never switch branches, never push anywhere else.
- Make no changes outside the named stage's scope. If you find something else, report it, don't fix it.
- After the change: run the production build, verify what the stage's acceptance criteria ask for, and start the dev server as a live preview so I can look at it.
- End with a report: what changed, what you verified and how, what you could NOT verify, and any decision you need from me. Never claim a visual check you did not perform.
- Note: this sandbox is rebuilt between sessions — only files inside the repo persist, so never rely on /tmp for anything that must last.
- Tell me when the stage is done and remind me which stage is next.

Confirm you have read the brief and are ready, then wait for my stage prompt.

S0 — SAFETY NET

STAGE S0 – SAFETY NET (items T1, T2, T3, T4). One stage only; do not start any other stage.

CONTEXT

- Repo: /home/user/portfolio-arena (React 19 + Vite 8 + Tailwind 3.4 portfolio). Branch: arena/01a10c65-portfolio-arena – commit to this branch only.
- Base commit: 85f12aa, 77 tracked files, clean tree. Full context in audit/PORTFOLIO-MASTER-BRIEF.md (Part A).
- There is currently NO .gitignore, NO error boundary, and no MotionConfig.
- The folder audit/ contains untracked documentation – leave it alone (do not commit it unless I say so).

TASKS

1. T1 – Create /home/user/portfolio-arena/.gitignore with entries for: node_modules/, dist/, .DS_Store, *.local, .env*, .vite/, coverage/, and qa/*.png (QA screenshots are throwaway). Create this file FIRST, before running any install command. Then verify with ``git status --porcelain --untracked-files=all`` that the tree shows only the intended new files.
2. T2 – Add a React error boundary:
 - New file src/components/ErrorBoundary.jsx: class component with `getDerivedStateFromError` + `componentDidCatch`, rendering a branded fallback in the site's own language (ink background, display font, mono micro-label, "Back to home" link + reload button). Keep it dependency-light; inline styles are acceptable so it works even if the CSS chunk failed.
 - Wire it in src/App.jsx so it wraps `<AnimatedRoutes />` (keep Nav and Footer alive around it), and so it resets when the route path changes (e.g. key on location.pathname or a reset handler).
 - Verify by temporarily throwing inside a route component on a scratch basis, confirming the fallback renders, then REVERT the temporary throw. Report the evidence.
3. T3 – In src/App.jsx wrap the app in `<MotionConfig reducedMotion="user">` from 'motion/react' (one import + one wrapper). This makes reveals, parallax and the Hero orbit respect prefers-reduced-motion, as DESIGN.md §8 already promises. Do not change any other motion config.
4. T4 – Delete dead imports and the unused prop:
 - src/components/Section.jsx: remove unused ``useSite`` import.
 - src/sections/About.jsx: remove unused ``Terminal``, ``Stagger``, ``StaggerItem``.
 - src/sections/Education.jsx: remove unused ``Stagger``, ``StaggerItem``.
 - src/sections/Hero.jsx: remove unused ``EASE``.
 - src/sections/ProjectsHome.jsx: remove unused ``ExternalLink``, ``MaskLines``.
 - src/pages/ProjectDetail.jsx: remove unused ``Parallax``.
 - src/components/Section.jsx: remove the unused ``labelledBy`` prop (check no caller passes it). Change nothing else in those files.

ACCEPTANCE CRITERIA

- ``git status`` shows only the intended changes; node_modules (if installed) is NOT tracked.
- ``npm run build`` succeeds with no new warnings.
- The error boundary renders the branded fallback when a component throws (evidence captured).
- Reduced-motion: with the OS/browser set to reduce motion, reveals/parallax do not animate (verify by code inspection at minimum; say so if you cannot emulate it).
- No visual or behavioural change to the site in normal conditions.

REPORT + COMMIT

- Commit message: ``S0: add .gitignore, error boundary, reduced-motion config; remove dead imports``.
- Report: files changed, verification method for each item, anything unverified, and confirm whether the dev server preview is running. Then say: "S0 done – next: S1 images truth pass."

S1 — IMAGES TRUTH PASS

STAGE S1 – IMAGES TRUTH PASS (item T24). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- ALL 16 project screenshots are in the WRONG folders. Everything currently in public/projects/bmw/ is really CODING NINJAS; everything currently in public/projects/coding-ninjas/ is really BMW. Two images (bmw-08, bmw-09) are referenced nowhere.
- The exact verified mapping, new file names, gallery order and new alt text are in audit/PORTFOLIO-MASTER-BRIEF.md §8. Use them verbatim – do not re-derive or invent.
- This stage is PURE file moves/renames + data edits. Do NOT re-encode, resize or convert any image in this stage (metadata and WebP are separate stages S2/S6).

TASKS

1. Move + rename the 16 files exactly as listed in brief §8.2, using `git mv` so history follows:
 - 9 files -> public/projects/coding-ninjas/coding-ninjas-01-hero.jpeg ... coding-ninjas-09-footer.jpeg
 - 7 files -> public/projects/bmw/bmw-01-hero-5-series.png ... bmw-07-search-footer.png
 - Result: public/projects/bmw/ contains ONLY BMW files; public/projects/coding-ninjas/ ONLY CN files.
 - No orphans: bmw-08 and bmw-09 (alumni-ratings, footer) become CN items 7 and 9.
2. Rewrite the `gallery` arrays in src/data/projects.js for both projects:
 - correct `src` paths and the new `alt` text from brief §8.3 (verbatim);
 - add a short `label` to each gallery item describing the section (used by the sticky story);
 - update `scrollFrames` for the Coding Ninjas project now that its gallery has 9 entries (keep the existing narrative intent: overview -> sections -> tech/status -> explore);
 - keep every other field (title, tech, github, status, description, sections/pages, implementation) unchanged.
3. src/sections/ProjectsHome.jsx: delete the hardcoded FRAME_LABELS array and read captions from data instead: project.gallery[frame].label (with a safe fallback if a label is missing).
4. Docs: update README.md (asset section) and UPDATE_GUIDE.md (step 2) to document the naming pattern `-NN-<content>.<ext>` and the rule "one gallery array per project, in page order; alt text must describe what is actually visible".
5. Keep the existing honesty labels unchanged ("Website Recreation", no affiliation).

ACCEPTANCE CRITERIA (verify all, show evidence)

- No file remains in the wrong folder; the two orphaned images are attached; count = 16 files.
- Every `gallery[].src` in projects.js exists on disk, and every image on disk is referenced exactly once (write a one-off check and show its output).
- Both project detail pages render their own screenshots in the intended order (/projects/coding-ninjas shows CN UI, /projects/bmw shows BMW UI). If you cannot run a browser, prove it by listing the resolved src order per project and by checking the alt text.
- `npm run build` succeeds; no other route or section changes appearance.

REPORT + COMMIT

- Commit message: `S1: swap misplaced project screenshots, rename to project-based names, re-point galleries + alt text`.
- Report: the mapping table as executed, the existence/reference check output, anything unverified. Then say: "S1 done – next: S2 metadata sweep."

S2 — METADATA SWEEP (ALL FILES)

STAGE S2 – METADATA SWEEP (user correction C2). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- BINDING USER INSTRUCTION (C2): remove metadata from EVERY file in the portfolio – all 7 PDFs, all 16 screenshots, and public/favicon.svg – not just the two PDFs that were found to carry personal data. Verified per file afterwards.
- Do not change any visible content: no re-compression of image data, no cropping, no re-rendering.
- Method (already tested on copies, see brief §6.5):
 - * PDF: pikepdf – clear the XMP metadata packet, delete the document Info dictionary, re-save with compress_streams=True. (pip install pikepdf if needed.)
 - * PNG: lossless re-save dropping ancillary chunks (tEXt/iTXt/zTXt/tIME/eXIf) – pixels identical.
 - * JPEG: re-save with quality='keep', subsampling='keep' so the entropy-coded image data is NOT recompressed; only APPn/COM metadata markers are dropped.
 - * SVG: strip XML comments and any <metadata>/<title>/<desc> nodes; keep the drawing byte-identical.
- S1 must already be done (file names are final).

TASKS

1. Record the BEFORE state of every file: for each of the 7 PDFs, 16 images and favicon.svg, extract whatever metadata exists (Info/XMP for PDFs, text/time/EXIF chunks for PNG/JPEG, metadata nodes for SVG) and print a table. Expect: resume.pdf (Author "Un-named", Word 2010, dates), ysf-internship-certificate.pdf (Author "Richa Garg", Canva), five certificates with tool-only metadata, images clean, favicon likely clean.
2. Strip metadata from all files with the methods above, in place, keeping the same file names.
3. Add a reusable verifier script that ENFORCES the result: qa/check-metadata.mjs (Node) or qa/check_metadata.py (Python) – whichever is cleaner to run without extra deps – that scans public/ recursively, prints a per-file verdict, exits non-zero if ANY metadata is found, and prints "0 files with metadata" on success. Wire it as an npm script: "check:meta": "node qa/check-metadata.mjs".
4. Integrity proof for every changed file: page count unchanged for PDFs, and for PDFs the page content-stream SHA-256 must be IDENTICAL before/after; for images, pixel data identical (compare decoded pixels or file hash of image data); for the SVG, the rendered paths unchanged. Show the before/after table.
5. Note in the report which files actually had metadata (so the user knows what was exposed) and which were already clean.

ACCEPTANCE CRITERIA

- `npm run check:meta` exits 0 and reports zero files with metadata across public/.
- Every PDF: same page count, identical content-stream hash.
- Every image: identical decoded pixels, same dimensions; no visible change.
- `npm run build` succeeds; site renders identically (gallery images and all certificate links still work).
- Nothing else in the repository was touched.

REPORT + COMMIT

- Commit message: `S2: strip metadata from all public assets (7 PDFs, 16 images, favicon) + add check:meta verifier`.
- Report: before/after metadata table, integrity proofs, verifier output, anything unverified. Then say: "S2 done – next: S3 404 + deep links."

S3 — 404, DEEP LINKS & FAILURE SURFACES

STAGE S3 – 404, DEEP LINKS & FAILURE SURFACES (items T21, T22). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena. Deploy target: VERCEL.
- Today: the in-app 404 route (src/pages/NotFound.jsx, path="*") works only inside React Router. There is NO public/404.html, NO SPA rewrite config, so on Vercel a direct load of /projects/bmw, /certifications, /contact etc. returns Vercel's own 404. And if JS fails to load, <div id="root"></div> renders nothing at all.
- The user explicitly asked for a proper custom 404 page. The three failure surfaces (host 404, React 404, error-boundary fallback from stage S0) must look like ONE design language.

TASKS

1. Add vercel.json at the repo root with a SPA rewrite:
{ "rewrites": [{ "source": "/(.*)", "destination": "/index.html" }] }
(Do not add "cleanUrls"/"trailingSlash" experiments.)
2. Add public/404.html – a self-contained, no-JS branded page that Vercel serves for unmatched URLs and that also covers the "JS never loaded" case:
 - inline CSS only (no external requests), ink background, ivory type, the site's mono micro-label style, "error 404 – route not found" copy in the spirit of the existing NotFound page ("This page never shipped."), plus a prominent "Back to home" link.
 - Must be readable at 320px width and pass a contrast check (ink #10162B on ivory text etc.).
 - No build step may be required for it: it must exist as-is in dist/ after `vite build` (public/ is copied verbatim).
3. Make the React 404 and the new host 404 consistent: same headline, same mono micro-label, same CTA wording. Adjust src/pages/NotFound.jsx only if needed for consistency (keep its current design quality and its existing copy unless a change is needed for the match).
4. Confirm the S0 error boundary's fallback also matches this visual language (adjust wording/spacing if it drifted).
5. Add a short note to README.md documenting the failure surfaces and how to test them locally: `npm run build && npm run preview`, then visit /nope (host 404) and /projects/bmw (deep link must load the case study, not 404).

ACCEPTANCE CRITERIA

- `npm run build` copies 404.html into dist/ (show it).
- With `npm run preview`: a deep link like /projects/bmw loads the case study; an unknown path shows the design-consistent 404; with JavaScript disabled the 404.html content is still readable. If you cannot run a browser, verify by inspecting the built files and the rewrite rule, and say clearly that the browser check is pending the user's eyes.
- No other route's behaviour changes; no visual change to the rest of the site.

REPORT + COMMIT

- Commit message: `S3: add Vercel SPA rewrites + branded no-JS 404 page; unify failure surfaces`.
- Report: files added, how the rewrite works, what was verified, what needs the user's browser. Then say: "S3 done – next: S4 bug fixes."

S4 — BUG FIXES (BEHAVIOUR)

STAGE S4 – BUG FIXES (items T14, T15, T16, T17, T18, T19 + A2 timeout part). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- BINDING CORRECTION C1: do NOT add any "Messages are delivered via FormSubmit (third party)" line, or any similar third-party disclosure, under the contact form. The user rejected that suggestion. You MAY still add the request timeout and fix the misleading error copy.
- Six verified defects to fix (details in audit/PORTFOLIO-MASTER-BRIEF.md §5, evidence in §4).

TASKS

1. T14 – navbar tone bug + duplicate sections/glows:
 - src/lib/site.jsx: make `useSectionSignal` restore the previous tone/accnt when a section LEAVES the middle band (track a stack or remember the last value from another intersecting section), so the nav can never get stuck in the wrong tone. Keep the API (tone, accent) -> ref unchanged for existing callers.
 - src/components/Section.jsx: add an optional `intro` slot (rendered above the children, inside the same <section>) so a page header can live inside the single section.
 - Rework the 5 wrapper pages in src/pages/SimplePages.jsx (Experience, Education, Certifications, Achievements, Contact) and src/pages/AboutPage.jsx so each page has EXACTLY ONE <Section> (with the PageIntro passed via `intro`) instead of a <Section> wrapping another <Section>. Preserve the existing background/accnt per page (ivory + the same accent colour) and the existing visual rhythm – this must not change how the pages look.
 - Also fix the `bare` props (About/Skills/ProjectsHome/Experience/Education/Certifications/Achievements/Contact) so nothing renders a second section or a stray SectionHead.
 - Verify with a static render: build an esbuild script that renders each route with react-dom/server and count <section> elements and radial-gradient glows – expect 1 and 1 per page. Show the before/after counts (before: /contact 2/2, /certifications 2/2, /about 2/2).
2. T15 – single scroll engine:
 - Create src/lib/scroll.jsx exposing the ONE Lenis instance via React context (provider + useLenis hook). src/App.jsx should create/destroy it there (still skipped under reduced motion).
 - Route change: replace `window.scrollTo({top:0})` with `lenis.scrollTo(0, { immediate: true })`.
 - Footer "Back to top": use `lenis.scrollTo(0)` (fall back to window.scrollTo when no Lenis, e.g. reduced motion).
 - Call `lenis.resize()` after each route renders so dimensions stay correct.
3. T16 – mobile menu:
 - src/components/Nav.jsx: while the overlay is open call lenis.stop() and on close lenis.start() (guard when Lenis is absent), keep the body overflow lock, add role="dialog" aria-modal="true" aria-label to the overlay, put the background content behind `inert`/aria-hidden while open, trap Tab focus inside the overlay, and return focus to the hamburger button on close (Escape already closes it).
4. T17 – cursor affordances:
 - src/sections/ProjectsHome.jsx: make the featured browser mockup a real <Link> to the project case study (it currently carries data-cursor="project" and does nothing).
 - src/sections/Certifications.jsx and src/sections/Experience.jsx (simulation cards): either wrap each card in a real (preferred – bigger tap target, and the "OPEN" cursor then tells the truth) or downgrade the cursor attribute on genuinely non-clickable elements. Do not leave a "OPEN"/"VIEW" cursor on something that does nothing.
 - Keep all existing hover/spotlight styling working.
5. T18 – Hero viewport:
 - Add src/hooks/useViewport.js (width/height state, updated on resize + orientationchange).
 - src/sections/Hero.jsx OrbitVisual: use it instead of reading window.innerWidth/innerHeight during render. The visual behaviour must be identical, but it must not freeze after a resize and must not throw when rendered outside a browser (verify: rendering <Hero> with react-dom/server no longer throws "window is not defined").
6. T19 + A2 – contact form robustness (src/sections/Contact.jsx):
 - Add an AbortController with a 10-second timeout to the fetch; on timeout/abort/network failure set the error state (button recovers, never hangs on "Sending...").
 - Fix the error copy that says "email me directly below" so it matches where the address actually is on the page.
 - Do NOT add the rejected third-party disclosure line (C1). Do not change the endpoint in this stage (bundle/privacy work is stage S7).

ACCEPTANCE CRITERIA

- Static render proof: exactly 1 section + 1 glow per route (show counts for all 7 routes).
- /contact and /certifications: scrolling to the bottom (dark content) and back to the top leaves the navbar in the correct tone. If you cannot run a browser, prove it by explaining the intersection logic and ask the user to confirm on the preview.
- One Lenis instance; no remaining direct window.scrollTo on route change or footer (grep to show it).
- Mobile menu: Lenis stopped while open, focus trapped, focus restored, inert background.
- Hero renders in SSR without throwing (show the command + output).
- Form: simulated slow/failed request recovers to the error state within ~10 s.
- `npm run build` succeeds; site otherwise visually unchanged.

REPORT + COMMIT

- Commit message: `S4: fix nav tone + duplicate sections, single Lenis scroll, mobile menu ally, cursor affordances, hero viewport, form timeout`.
- Report per item with evidence, anything unverified, then: "S4 done – next: S5 accessibility."

S5 — ACCESSIBILITY

STAGE S5 – ACCESSIBILITY (items T25, T26, T5, T7, T12). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- Five WCAG-AA contrast failures were MEASURED. The replacement values in audit/PORTFOLIO-MASTER-BRIEF.md §6.2 are already computed and verified – use them verbatim, do not invent new colours and do not re-derive.
- Goal: meet AA without changing the art direction ("Signal & Ink"). These are small darkening steps, not a redesign.

TASKS

1. T25 – contrast fixes:
 - a) tailwind.config.js: `muted: '#667085'` -> `'#5B6474'`.
 - b) tailwind.config.js: add `cobalt-ink: '#1D4ED8'` (for accent text on LIGHT backgrounds only; keep bright `cobalt` for dark backgrounds).
 - c) src/index.css base layer: focus ring – light sections use an ink ring, dark sections keep cyan. Simplest correct approach: default `:focus-visible { outline: 2px solid #10162B; outline-offset: 3px; }` and for `.dark-zone :focus-visible { outline-color: #27D3F2; }` (remove/replace the current ring-cyan utilities so the ring is always visible). Verify contrast: ink on ivory = 16.03:1, cyan on deep = 9.65:1.
 - d) Primary CTA buttons that are indigo with ivory text (Hero "View projects", Nav shell, anywhere `bg-indigo` + `text-ivory`): change the resting background to `#5A4BD4` (hover can stay `#5a4bd4` or go slightly deeper). Ivory on #5A4BD4 = 5.54:1.
 - e) src/sections/Contact.jsx inputs: border `border-ivory/20` -> `border-ivory/45`; placeholder `placeholder:text-fog/50` -> `placeholder:text-fog/80`.
 - f) src/sections/Achievements.jsx CHIP coordination: `text-[#0e7d90]` -> `text-[#0b6b7c]`.
 - g) Any `text-muted/80` micro-note on light backgrounds -> full `text-muted` (do not use opacity).
 - h) Where small accent text sits on light backgrounds, prefer the new `cobalt-ink` (e.g. src/sections/Experience.jsx simulation labels, src/pages/SimplePages.jsx). Coral is decorative only on light backgrounds: keep it for icons/marks that are not the sole carrier of information, and darken where it is text.
2. T26 – heading order: every route must produce h1 -> h2 -> h3 without skipping a level.
 - Static render shows /experience, /education, /certifications, /achievements, /contact currently jump from h1 straight to h3 (their page intro is an h1 and the section head is absent because those pages render "bare").
 - Fix by rendering the section heading on those pages (preferred – reuses SectionHead, which is an h2 with a proper id) OR by demoting the entry titles to h2 when no section head is present. Keep exactly one h1 per page and keep the visual design identical.
 - Verify with a static render that prints the heading outline per route.
3. T5 – src/components/Marquee.jsx: the duplicate row must be hidden from assistive tech. Currently only the second row carries aria-hidden. Hide BOTH rows from the ally tree and expose the words once for screen readers (an sr-only list, or aria-hidden on the visual rows plus a single sr-only sentence). Do not change the visual animation.
4. T7 – src/pages/SimplePages.jsx AchievementsPage: the PageIntro accent is coral while the section is lime. Align the page intro accent with the section accent (lime) so the one-accent rule holds. (If you think coral was intentional, report instead of changing – but the design doc says one accent.)
5. T12 – unguarded lookups: add safe fallbacks so a data change can never break the UI:
 - src/sections/Achievements.jsx: TILE[item.id] / CHIP[item.id] -> fall back to a sensible default tile/chip style and log nothing in production.
 - src/sections/Skills.jsx: accentText[g.accent] / hoverBg[g.accent] -> fallback.
 - src/sections/Certifications.jsx: accentText[cert.accent] / tileBorder[cert.accent] -> fallback.
 - src/components/SectionHead.jsx: accentBg[accent] -> fallback.Prove it: temporarily add a fake achievement/skill with an unknown id/colour, confirm the page still renders, then revert the temporary data.

ACCEPTANCE CRITERIA

- Re-measure and report the contrast ratios after the change (at minimum: muted on ivory/tint/paper, ivory on the new button background, focus ring on light and dark, form border + placeholder, teal chip). All must meet 4.5:1 for text and 3:1 for UI/focus.
- Heading outline per route: exactly one h1, no skipped levels (show the outline).
- Marquee: screen reader text appears once (show the DOM).
- Unknown ids/colours render with fallbacks (evidence captured, temporary data reverted).
- DESIGN.md: update the two incorrect contrast claims if S9 is not yet done (muted and indigo numbers) so the doc matches reality – otherwise leave it for S9 and say so.
- `npm run build` succeeds; visual diff is limited to the small darkening of secondary text/buttons.

REPORT + COMMIT

- Commit message: `S5: meet WCAG AA (focus ring, muted, cobalt-ink, CTA, form borders, chips), heading order, marquee ally, guarded lookups`.
- Report the measured before/after table, the heading outlines, anything unverified. Then say: "S5 done – next: S6 performance."

S6 — PERFORMANCE

STAGE S6 — PERFORMANCE (items T31, T32, T30, T10). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- Measured baseline (production build): JS 519.81 kB in ONE file (gzip 160.26, Vite warns >500 kB), CSS 50.39 kB (gzip 14.44), 14 woff2 = 214 kB, dist = 12 MB.
- Pre-tested WebP result (brief §6.4): lossless WebP is a 49% win for the 7 PNG-backed BMW files and a 95% LOSS (2x bigger) for the 9 JPEG-backed Coding Ninjas files. So: convert ONLY the PNG group. For the JPEG group the default is to keep JPEG; only use lossy WebP if the user has approved it (ask, do not assume).

TASKS

1. T31 — route-level code splitting:
 - Convert route components in src/App.jsx to React.lazy + <Suspense> (keep Home eager so the hero paints immediately), with a lightweight fallback that matches the site (no blocking loader).
 - Add a `build` section to vite.config.js: target 'es2020', explicit chunk naming, chunkSizeWarningLimit raised only if justified, and keep sourcemaps off for production (or 'hidden' if the user prefers).
 - Report the new bundle table (per-chunk kB + gzip) and state clearly how much first-load JS the home route now needs versus before (519.81 kB / 160.26 kB gzip baseline).
 - Verify every route still loads (static render or preview navigation) and that the route transition animation still works.
2. T32 — images + fonts:
 - Convert the 7 BMW screenshots (currently .png) to lossless WebP with libwebp (Pillow: `save(..., 'WEBP', lossless=True, quality=100, method=5)`), keep the ORIGINAL .png files in the repo as fallback sources, and update the data to the WebP paths.
 - Implement the images with <picture> (WebP source + original fallback) and add `srcset` at two widths (e.g. 700w and 1400w) generated from the originals; add explicit width/height attributes (dimensions are listed in brief §6.3) to remove layout shift.
 - Coding Ninjas (JPEG-backed): keep JPEG, but add width/height + `decoding="async"` + `loading="lazy"` (lazy already exists in places — make it consistent below the fold only).
 - Fonts: import only the latin subsets needed. Current imports in src/main.jsx pull all subsets of the three variable fonts (14 files, 214 kB). Import the latin-only files and preload the two fonts used above the fold in index.html. If a subset file for a language is genuinely needed, keep it — otherwise drop.
 - Optionally (ask first, show a before/after legibility comparison): recompress the images INSIDE ysf-internship-certificate.pdf — it is 4.43 MB of embedded imagery; metadata work is already done in S2 and does not shrink it.
3. T30 — aspect fix for the sticky story: the CN screenshots are ~2.2:1 in an `aspect-[16/10]` box with object-cover object-top. Ask the user whether to retune the story viewport for the wide shots (e.g. aspect-[2/1]); implement only if they agree.
4. T10 — src/sections/ProjectsHome.jsx: guard the featured-project GitHub button so it does not render `href={null}` when a project has no repo (render the "source not published" style instead).

ACCEPTANCE CRITERIA

- `npm run build` output table before/after; first-load JS reduced substantially (report the exact numbers).
- Every route renders after the split (no blank screens, no console errors).
- Images: same visual appearance, correct paths, width/height present, no layout shift (verify by inspecting rendered HTML for width/height and by the file sizes).
- WebP savings table (per file + total) and confirmation that originals are still in the repo.
- Font payload reduced (report the number of font files shipped and total kB).
- `npm run check:meta` still passes (S2 verifier) — new/derived assets must not reintroduce metadata.

REPORT + COMMIT

- Commit message: `S6: route code splitting, lossless WebP + srcset for BMW screenshots, latin-only fonts, image dimensions, guard featured repo link`.
 - Report: before/after tables (JS, fonts, images, total dist), anything unverified, any decision pending.
- Then say: "S6 done — next: S7 discoverability."

S7 — DISCOVERABILITY & PRIVACY

STAGE S7 – DISCOVERABILITY & PRIVACY (items T27, T28, T29, T20). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena. Deploy target: Vercel.
- Today: no og:image / twitter:image / og:url / canonical (a shared link shows no preview), and Home calls usePageMeta(null, ...) which OVERWRITES the carefully written static title+description from index.html. No robots.txt, no sitemap.xml, no manifest. The contact recipient Gmail is baked into the public JS bundle.

TASKS

1. T27 – social + canonical:
 - Add to index.html: og:url, og:image (+ og:image:width/height/alt), twitter:image, twitter:card (summary_large_image), and <link rel="canonical">.
 - Generate public/og-cover.png at 1200x630 in the site's own design language (ivory/ink, the wordmark, the headline "Computer Science student building with Software, Web & Generative AI", indigo/cyan accents). It must be a real asset, not a placeholder. If you cannot generate imagery in the environment, say so and provide the exact construction instructions instead of shipping a broken tag.
 - Stop Home from clobbering the static meta: src/pages/Home.jsx / src/hooks/usePageMeta.js should leave the index.html title+description intact on '/'.
2. T28 – public/robots.txt (allow all, point to the sitemap) and public/sitemap.xml listing the 9 real routes with a single canonical origin placeholder that the user must set (flag it clearly).
3. T29 – public/site.webmanifest (name, short_name, description, theme_color #F6F2E8, background_color, display standalone, icons) + an Apple touch icon (180x180, derived from the favicon design, ink background + the wordmark) + the matching <link> tags in index.html.
4. T20 – reduce the exposed email in the bundle:
 - src/sections/Contact.jsx: move the endpoint behind an env-driven config (import.meta.env.VITE_CONTACT_ENDPOINT) with a safe fallback, and switch to FormSubmit's alias/hashed endpoint form so the literal address does not appear in the built JS.
 - BINDING CORRECTION C1 still applies: do NOT add any third-party disclosure line under the form.
 - Add .env.example documenting VITE_CONTACT_ENDPOINT (and ensure .gitignore from S0 already covers .env*).
 - Verify by grepping the built bundle: `grep -r "ashish1492a@gmail" dist/` must return nothing.

ACCEPTANCE CRITERIA

- Built dist/index.html contains the new meta tags; canonical + og:image resolve to real files.
- robots.txt + sitemap.xml are present in dist/ and syntactically valid.
- Manifest + apple icon present and linked; Lighthouse-style checks would not error on them.
- `grep -r "ashish1492a@gmail" dist/` returns nothing while the form still posts successfully (test the request path or explain exactly how to activate the alias).
- Home route keeps the rich static title/description (show the rendered <title> + meta for '/').
- `npm run build` succeeds; nothing else changes visually.

REPORT + COMMIT

- Commit message: `S7: social preview + canonical, robots/sitemap, web manifest + apple icon, move contact endpoint out of the bundle`.
- Report: files added/changed, the bundle grep proof, the canonical origin the user must set, anything unverified. Then say: "S7 done – next: S8 positioning."

S8 — POSITIONING: FULL STACK DEVELOPMENT AS EXPERTISE (binding user correction C3)

STAGE S8 – POSITIONING: FULL STACK AS EXPERTISE (user correction C3). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- BINDING USER INSTRUCTION (C3): the site must present FULL STACK DEVELOPMENT as an EXPERTISE AREA, not as something being learned. Today the site never claims it, while the user's CV headline lists "Full Stack Development" – the two contradict each other, and the site docs even say learning topics are never shown as expertise. C3 resolves that in favour of expertise.
- HARD LIMIT: do not invent technologies, frameworks, years, metrics, companies or counts. The Full Stack group may only contain technologies already present in src/data (frontend: HTML, CSS, JavaScript, React, Tailwind CSS, Vite; server-side & data: Python, Java, SQL, DBMS/RDBMS; tooling: Git, GitHub). Do NOT name Node/Express/Django/Flask unless the user confirms it – ask if the claim needs a named backend framework and wait for the answer before adding it.
- Keep the third-party protections EXACTLY as they are: "Website Recreation" labels, "Virtual Job Simulation – not employment", and the "no affiliation/endorsement implied" wording. Those are about other companies, not about self-assessment, and must not be weakened.

TASKS

1. src/data/skills.js: add a new skill group "Full Stack Development" (built only from the technologies above), with a short honest `note` (e.g. "End-to-end: interface, logic and data – project-built"), an accent from the existing set, and renumber the group indexes so the displayed order stays sequential (01...07). Do not touch the `coursework` list.
2. src/data/profile.js: reflect the positioning where it is honest and consistent:
 - add a `facts` entry or adjust an existing one so the hero micro-facts mention Full Stack Development (do not exceed four facts);
 - keep `headline` unless you propose a change and the user approves it (report as a proposal, do not apply silently);
 - keep the `currentlyExploring` chips for infrastructure topics (Linux, Docker, hosting, APIs) as they are – C3 was about full stack, not about removing the learning framing everywhere.
3. src/sections/About.jsx and/or src/sections/Skills.jsx: make the positioning legible – the skills section must show Full Stack Development as a first-class group, and the About copy may name it. Keep the existing visual language (no new components).
4. Docs consistency (this is part of C3, not optional):
 - README.md: update the "Honesty policy baked into the content" section so it no longer says learning topics are never presented as expertise; state instead that Full Stack Development is claimed as an expertise area, while the recreation labels, the simulation labels and the no-affiliation statements remain in force.
 - UPDATE_GUIDE.md: same correction, in the checklist wording.
 - DESIGN.md: no change needed (it is about visuals) – but if it repeats the honesty claim, align it.
5. Report the exact final skill-group order and the wording you used, so the user can approve the claim.

ACCEPTANCE CRITERIA

- Skills section renders 7 groups with sequential indexes, Full Stack Development among them, containing only technologies already present in the data.
- No invented technology, framework, year, metric or employer anywhere.
- README.md and UPDATE_GUIDE.md contain no statement that contradicts the new positioning.
- Recreation/simulation/no-affiliation labels untouched (show a diff summary proving they are intact).
- `npm run build` succeeds; layout unchanged apart from the new group.

REPORT + COMMIT

- Commit message: `S8: present Full Stack Development as an expertise area; align README/UPDATE\_GUIDE honesty wording`.
- Report: the new group content, the final index order, doc changes, the open question about naming a backend framework, anything unverified. Then say: "S8 done – next: S9 hygiene & content."

S9 — HYGIENE, CONTENT & TESTS

STAGE S9 – HYGIENE, CONTENT & TESTS (items T6, T8, T11, T13, T23, T33, T34). One stage only.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena.
- ASK FIRST (do not invent): the user must supply the year for the badminton achievement (T6). If they have not supplied it by the time you reach it, skip that single item, note it, and continue.

TASKS

1. T6 – src/data/achievements.js: add the missing year to the badminton entry (use only the year the user provides; if none, leave it out and report).
2. T8 – cursor/design mismatch: DESIGN.md §7 says the rail cursor label is "SCROLL →" while src/components/Cursor.jsx uses "SWIPE →". Ask the user which one wins, or (default) make the code match the design doc. State what you did.
3. T11 – remove stale hardcoded copy:
 - src/pages/ProjectsPage.jsx "Two recreations, one obsession" -> derive the count from the data (e.g. `\${projects.length} recreations`) without breaking the editorial line break;
 - src/sections/ProjectsHome.jsx "Project 02" -> derive the index from the data;
 - any other hardcoded project/section count you find while doing this.Prove it: temporarily add a third project to the data, confirm the copy updates, then revert.
4. T13 – DESIGN.md corrections: the muted contrast claim ("5.0:1" -> the measured value after S5) and the indigo claim ("4.6:1" -> measured). Also fix or implement the claim that Lenis is paused when the tab is hidden (either implement a visibilitychange pause or correct the document – recommend implementing it, it is a few lines and matches the doc).
5. T23 – qa/shoot.mjs: wire it to npm as `qa`: "node qa/shoot.mjs", replace the hardcoded `/home/user/qa` output path with a path inside the repo (e.g. qa/screens/ – already gitignored in S0), and fix the overflow check: `document.documentElement.scrollWidth - window.innerWidth` is defeated by `body { overflow-x: clip }`. Measure overflow on `document.body.scrollWidth` (or compare against `document.body.getBoundingClientRect().width`) so real horizontal overflow is actually detected. Add the BMW project detail page to the shoot list so the S1 mix-up class of bug is caught.
6. T33 – tooling: add ESLint (flat config, with react + react-hooks plugins) and Prettier, plus scripts `lint`: "eslint .", `format`: "prettier --write .". Run lint once and fix only real errors – do not reformat the whole codebase in this stage (that would bury the diff). Report the lint output.
7. T34 – Playwright smoke tests in tests/smoke.spec.js (and `test`: "playwright test"):
 - every route renders without console errors;
 - no horizontal overflow at 320/375/768/1440;
 - each project detail page shows ITS OWN screenshots: assert that the images on /projects/coding-ninjas resolve under /projects/coding-ninjas/ and the same for /projects/bmw (this is the regression guard for the S1 fix);
 - the 404 route and the host 404 render the branded page.Note: browser downloads may be unavailable in the environment – if so, still commit the tests, document how to run them (`npx playwright install chromium && npm test`), and say clearly that they were not executed.

ACCEPTANCE CRITERIA

- `npm run lint` runs and reports no errors (warnings allowed, listed).
- `npm run qa` and `npm test` exist and are documented in README (how to install browsers and run).
- The stale-copy proofs are captured and the temporary data is reverted.
- DESIGN.md no longer contradicts the implementation (contrast numbers + the Lenis claim).
- `npm run build` succeeds; no visual change.

REPORT + COMMIT

- Commit message: `S9: content nits, DESIGN.md corrections, npm-wired QA + overflow fix, ESLint/Prettier, smoke tests`.
- Report: each item, what was verified versus not executed, the open badminton-year question. Then say: "S9 done – next: S10 Tailwind 4 (isolated, last)."

S10 — TAILWIND 4 MIGRATION (ISOLATED, LAST)

STAGE S10 – TAILWIND 4 MIGRATION (item T35). Run only after the user has committed S0–S9 to main. One stage only. This is the only breaking change in the programme.

CONTEXT

- Repo: /home/user/portfolio-arena. Branch: arena/01a10c65-portfolio-arena (or a fresh branch off the user's updated main if they ask – follow their instruction).
- Reason for this stage: `npm audit` reports 5 high advisories (braces, chokidar, micromatch, fast-glob, tailwindcss) – all dev/build-only, all in the Tailwind 3 tooling chain. Shipped dependencies are clean.
- Tailwind 4 removes that chain, which is the only way to clear the advisories.
- This is a config-layer migration, NOT a rewrite: class names in JSX stay. Scope:
 - * src/index.css: `@tailwind base/components/utilities` -> `@import "tailwindcss";`
 - * tokens: either move tailwind.config.js into CSS `@theme { ... }`, or keep the JS config and load it with `@config "./tailwind.config.js";` (the lower-risk compatibility path – choose this unless the user asks for the modern form)
 - * postcss.config.js + package.json: `@tailwindcss/postcss` (or the `@tailwindcss/vite` plugin)
 - * custom utilities (bg-grid-light, bg-grid-dark, bg-dots-dark, noise-layer, mask-line, spotlight, underline-slide, edge-fade-x) -> `@utility` or plain CSS in a layer
 - * audit for v4 renames: default border colour becomes `currentColor`, default ring width `3px` -> `1px`, `shadow/rounded` renames, `bg-opacity-*` -> slash syntax.

PROCEDURE (follow in order, report after each step)

1. Before touching anything: `npm run build` and record the CURRENT bundle numbers + take the QA screenshots (qa/shoot.mjs) at 375/768/1440 for every route. These are the "before" baseline.
2. Create the migration on a branch (or on the session branch if the user says so). Commit the config migration separately from any follow-up fixes so a revert is surgical.
3. Migrate per the scope list. Then `npm run build` and compare bundle numbers.
4. Fix regressions one at a time; after each, re-run the build and the QA screenshots.
5. Produce a before/after table: routes checked, anything that changed visually (borders, rings, shadows are the likely candidates), and the bundle delta.
6. `npm audit` must now report 0 vulnerabilities in the tailwind chain (show the output).
7. If ANY unresolved visual regression remains, STOP, report it with evidence, and recommend `git revert` of the migration commit rather than shipping it.

ACCEPTANCE CRITERIA

- `npm audit` shows no advisories in the tailwind/dependency chain (or only ones unrelated to it, listed).
- Every route renders identically to the before-screenshots (the user confirms the visuals; state plainly that your verification is code + build + their screenshots).
- `npm run build` succeeds; bundle size is not worse.
- `npm run lint`, `npm run check:meta` and the smoke tests still pass.
- One isolated commit, revertible without touching any other stage.

REPORT + COMMIT

- Commit message: `S10: migrate Tailwind 3.4 -> 4 (config layer), clears 5 dev-only advisories`.
- Report: the before/after table, the audit output, anything the user must eyeball, and an explicit statement of what a revert would need to undo. Then summarise the whole programme's status (which stages are done, which remain).

End of Part B. Keep this file together with [PORTFOLIO-MASTER-BRIEF.md](#) in `audit/`.